

Title: From Attitudes to Behaviour: How Math Attitudes Influence Students' Behavioural Choices in Primary School

Commentato [AM1]: è solo una proposta

Study Information

For details about study design and method, see the section below.

Hypotheses

The aim of this study is to investigate the following research questions, for which statistical power is planned.

The first research question (RQ1) is addressed to determine the most appropriate statistical treatment of the outcome variable for the primary analyses (RQ2). Beyond its statistical relevance, RQ1 also carries theoretical importance: establishing the underlying structure of behavioural responses related at individual trait-level constructs (such as avoidance and buoyancy) is a prerequisite for evaluating whether such constructs are meaningfully captured by the observed behaviour.

RQ1: Do students' behavioural choices during a mathematical task (i.e., three task-related choice: self-reliance, help-seeking, and skipping) reflect an underlying ordered continuum (i.e., an ordinal variable), or do they represent qualitatively distinct behavioural categories (i.e., categorical variable)?

HP1: If the three behavioural choices reflect an ordered dimension, they might be conceived as a continuum ranging from avoidance behaviour, from one side, to increasing of task engagement, on the other. Specifically, self-reliance behaviour reflects a tendency to solve the task independently, help-seeking behaviour reflects an attempt to engage with the task with external support, and avoidance behaviour indicates a reluctance to engage with the task. If no underlying order is supported at the measurement level, students' behavioural choices will instead be treated as categorical. Model comparisons between ordered and multinomial specifications will be used to determine which conceptualization best fits the data.

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RQ2: To what extent do trait-and-state level math attitudes (e.g., buoyancy and worry) predict students' behavioural choices (i.e., skipping, help-seeking, self-reliance behaviours) during the mathematical task?

Trait-level attitudes include math-related academic buoyancy, avoidance, anxiety, and self-efficacy; state-level measures will be collected for perceived self-competence and worry.

HP2: After establishing whether students' behavioural choices are best conceptualized as ordered or categorical (RQ1), we will examine how trait- and state-level math attitudes predict variability in these behaviours. Grounded in converging evidence (Choe et al., 2019; Putwain et al., 2020; Wang et al., 2025) we expect the following patterns:

The analyses will include four predictors in total: buoyancy, avoidance, worry, perceived competence; while others will serve as covariates (math anxiety and self-concept). All associations will be tested controlling for the relevant covariates, also including math ability and gender.

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Commentato [RP2]: Self-efficacy è sbagliato, self-competence sarebbe corretto, ma forse si confonde con stato?

Ecco le mie due proposte:

- self-concept (tratto) e perceived competence (stato)

- self-competence (tratto) e perceived competence (stato)

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Specifically, for the included trait-level measures we expect:

- Higher buoyancy → more self-reliant behaviour; lower buoyancy → more skipping (Putwain et al., 2020);
- Higher avoidance → more skipping; lower avoidance → more self-reliant behaviour (Choe et al., 2019; Wang et al., 2025)
- Moderate levels of buoyancy and lower avoidance → help-seeking (Wang et al., 2025)

As concerning state-level measures, we hypothesize:

- Higher perceived competence → more self-reliant behaviour; lower perceived competence → more skipping
- Higher worry → more skipping; lower worry → more self-reliant behaviour
- Moderate levels of both worry and perceived competence → help-seeking

Students may skip items either due to avoidance or metacognitive awareness of their limited ability, especially on more difficult items. To disentangle these processes, prior math ability from T0 will also be included as a covariate in all models to distinguish ability-driven from attitude-driven variation in choices.

Overall, if behavioural choices reflect an ordered continuum, predictors are expected to show linear effects consistent with increasing levels of task engagement. Conversely, if behaviours represent distinct categories, predictors are expected to differentially predict specific behavioural choices without implying a single common underlying dimension.

Design Plan

Study type

Behavioural study – Participants will complete two small-group sessions of a computerized task in a controlled setting

Blinding

Is there any additional blinding in this study?

Study design

The study employs two sessions (T0 and T1) to examine the relationships between math attitudes (both trait- and state-level), math behavioural choice, and math ability.

In the first session (T0), participants complete 16 arithmetic problems, followed by the two state questionnaires (worry and perceived competence). After other 16 arithmetic problems, participants are asked to answer two trait questionnaires (math anxiety and buoyancy). In the second session (T1), participants complete a set of 32 arithmetic problems with 3 possible choices (skip, ask for help and solve independently). After the math task, participants are asked to answer two questionnaires (avoidance and self-competence).

All tasks are administered in small groups via computer in the classroom.

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For both RQ1 and RQ2, the primary outcome variable is students' item-level behavioural choice during the math task (i.e., skipping, help-seeking, or self-reliance behaviour). Depending on the results of RQ1, this outcome will be treated either as an ordinal or a categorical variable. Trait- and state-level math attitudes will be included as predictors and will vary within participants. Behavioural choices will be analysed using regression models appropriate to the outcome structure (i.e., ordered or multinomial), while accounting for the repeated-measures nature of the data.

No special actions will be taken by the experimenter for participants showing extreme patterns (e.g., skipping all items). Such behaviour is discouraged by two task design features: (1) a mandatory 15-second dwell time per item after each choice, and (2) automated prompts every 5 consecutive skips or help requests, reminding participants of all the available behavioural choices.

Randomization

Sampling Plan

Existing Data

Data collection is ongoing (T0). Registration prior to any human observation of the data

Explanation of existing data

No data

Data collection procedures

Participants will be recruited by initially contacting the Heads of Public Primary Schools in middle-income areas of a Western European country. Once permission is granted by the teachers, parental consent will be requested. The study will involve students in the 4th and 5th grades, aged 9 to 10 years. Exclusion criteria include formal diagnoses of intellectual disabilities or neurological/genetic conditions. No financial compensation will be provided. The study has received approval from the Authors' University ethics committee for psychological research.

Data collection will take place during the second half of academic year (2025-2026). The first session (T0) will be conducted starting in early March 2026, followed by the second session (T1), which will be conducted between April and June 2026.

Sample size

The sample size is partially constrained by available resources. We aim to recruit approximately 35-40 classes, corresponding to an initial target of about 500 students. We anticipate a missing rate of approximately 8% per session, due primarily to school absences (on average, 1-2 out of 20 children per class on a given testing day). Additionally, we estimate that approximately 15-20 children per class will meet the inclusion criteria and return the signed parental informed consent. Based on these factors, we expect to include 400 eligible participants in the final analysis. Recruitment will continue until 400 eligible participants are obtained or all recruited classes complete both sessions, whichever occurs first. If resources limit recruitment below 400, analysis proceeds with the achieved $N \geq 300$, where simulations confirm $> 90\%$ power (ordinal conceptualization) or good posterior SEs for all primary parameters (categorical conceptualization) at $N=300$.

Sample size rationale

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Two separate power analyses were conducted to reflect the two possible conceptualizations of math-related behavioural choices tested in RQ1 (ordinal vs. categorical). Full details and the power analysis code are provided in the attached files “PowerAnalysisOrdinal.R” (ordinal conceptualization) and “FastPowerCategorical.R” (categorical conceptualization). In both cases, the power analyses are fully aligned with the planned statistical models outlined in the “Analysis Plan section” and are based on data-generating processes consistent with the large simulated dataset attached in “Study Design section”. The following paragraphs provide a concise overview of both power analysis procedures and their results.

1) Power analysis for the ordinal conceptualization of the dependent variable (math-related behavioural choices) was evaluated using a simulation-based approach with cumulative link mixed models (CLMM). The power analysis implemented in “PowerAnalysisOrdinal.R”, was based on a power simulation with 5,000 iterations Monte Carlo simulation iterations of the full model, including predictors relevant to both RQ1 and RQ2.

The simulated logistic linear regression model tested the effects of standardized predictors (buoyancy, avoidance, worry, perceived competence) on ordered behavioural choices collected at T1 (skipping < help-seeking < self-reliance), while controlling for all relevant covariates and random intercept, thus accounting for intersubjective variability. Under the alternative hypothesis, based on coefficients typically reported in literature, *betas* were set to -0.25 for state worry, -0.30 for trait avoidance, 0.30 trait buoyancy, and 0.25 for state self-competence, and a subject-level random intercept variance of $\tau = 1$, reflecting moderate individual differences between subjects **in baseline behavioural choice propensity**. These values correspond to statistically small effects, consistent with those typically observed in psychological research (odds ratios ≈ 0.78 for worry/avoidance, ≈ 1.35 for buoyancy/self-competence per 1 SD), equivalent to an effect size of Cohen's $d \approx 0.17$ using standard logistic conversion formulas.

The simulation incorporated the planned multilevel structure of the study, clustered by class/grade (20 classes $\times$ 20 students, aged 9-11 years), nested items (64 items total, 32 per session with session 2 as outcome), and realistic trait-state correlations via a multivariate covariance matrix. Results for H1 indicated power exceeding 80% for detecting each individual predictor effect at $\alpha = 0.05$ (directional, one-sided tests) at $N=300$. Joint power for detecting all four predictors significant simultaneously was approximately 83% without correction, remaining at $\sim 83\%$ after false discovery rate (FDR) correction across the four predictors (buoyancy, avoidance, worry, perceived competence).

In order to prove that the simulated process is valid and well-calibrated, null hypothesis (H_0) simulations have been conducted with all β s = 0, confirming Type I error rates near 5%. This result demonstrated no true effects present under the null hypothesis scenario, confirming no inflation of false positives.

2) Power analysis for the categorical conceptualization was conducted using simulation-based methods with multilevel multinomial logistic regression (brms). Due to excessive computational demands for the full model, a streamlined power analysis was implemented. The power analysis implemented in “FastPowerCategorical.R” simulates data across a grid of sample sizes ($N=100$ to

500 in steps of 50), averaging 10 replications per N to compute variance-covariance matrices of key posterior parameters for RQ2.

The simulated model tested the effects of standardized predictors (buoyancy, avoidance, worry, and perceived competence) on categorical behavioural choices at T1 (skip, help-seeking, self-reliance), while controlling for prior math ability at T0, trait math anxiety, trait self-competence, gender, and random intercepts. Under the alternative hypothesis (H1), logit coefficients for key contrasts were set to reflect plausible shifts in choice probabilities For skip vs. (help-seeking/self-reliance): buoyancy $\beta = -0.30$, avoidance $\beta = +0.30$, worry $\beta = +0.25$, perceived competence $\beta = -0.25$. For help-seeking vs. self-reliance: smaller effects in all the considered predictors (e.g., buoyancy $\beta = -0.15$). These yield odds ratios which will range ≈ 0.74 -1.35 per 1 SD across primary contrasts, equivalent to an effect size of Cohen's $d \approx 0.17$ using standard logistic conversion formulas. As before, the simulation incorporated the planned multilevel structure of the study clustered by class/grade (20 classes $\times$ 20 students, aged 9-11 years), nested items (64 items total, 32 per session with session 2 as outcome), and realistic trait-state correlations via a multivariate covariance matrix.

Power analysis prioritized the four primary contrasts central to HP2 (effects of buoyancy, avoidance, worry, and self-competence on self-reliance vs. skipping behaviour). Secondary help-seeking parameters were monitored for completeness, focusing especially on one additional parameter (buoyancy effect on help-seeking).

To determine a plausible sample size, we first identified desired posterior standard errors (SEs) for reliable estimation of primary effects after Bonferroni correction across five predictors, corresponding to 80% power. Across a simulation grid (N=100-500), posterior SEs for the four primary parameters fell below target thresholds (ranging from 0.035-0.079) by N=300: effects of buoyancy (0.033), avoidance (0.016), self-competence (0.032), and worry (0.067).

The secondary parameter (buoyancy effect on help-seeking) remained slightly elevated (0.068) compared to its stringent target (0.035) due to its smaller effect simulated ($\beta = 0.15$) and the conservative Bonferroni correction across five predictors, which is known to reduce power for smaller effects.

Further details on these assumptions and the complete R code for the power analysis can be found in the attached files "PowerAnalysisOrdinal.R" and "FastPowerCategorical.R".

Stopping rule

No data

Variables

Manipulated variables

Measured variables

The experiment includes three main set of measured variables: math performance, math behavioural choices, trait- and state-level math attitudes.

Math Performance: will be assessed in both sessions (T0 and T1) using two parallel digital algebraic tasks each consisting of 32 different problems (e.g., $34 - ? = 21$). A pool of 100 arithmetic problems

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will be generated using a custom R script, subject to several constraints (e.g., operands ranging from 12 to 99, exclusion of multiples of 10, and balanced positioning of the unknown term across items). The generated pool will be subsequently reviewed and manually adjusted.

For each participant, 32 problems per session will be randomly sampled from this pool. To prevent repetitions, the sampling algorithm will exclude problems already presented in T0 when drawing items for T1. For each item, students will have a time limit of 25 seconds, determined through a prior pilot test, to solve the problems and enter the correct answer using the keyboard. 1 point will be assigned for correct and timely answers, while 0 points for incorrect, late or skipped responses.

Math behavioural choices: will be assessed only in the second session (T1). For each item, participants will select one of three task-related options: (1) not solving the problem, indicating skipping behaviour; (2) requesting a hint, that provides a brief problem-solving strategy (e.g., for “ $34 - 16 = ?$ ” the hint would be “Try using the strategy: $34 + 16 = ?$ ”), indicating help-seeking behaviour; or (3) solving the problem independently, reflecting self-reliance behaviour.

For each item, the selected behavioural choice will be coded as 1 (logit), and the two non-selected options will be coded as 0, yielding an objective measure of students’ observable behavioural choices adapted from the paradigm proposed by Wang et al. (2025).

Trait measures will be assessed using the following self-report questionnaires:

- Math anxiety: will be measured using the Abbreviated Mathematics Anxiety Scale for Children (AMAS, Caviola et al., 2017). The 9-item measure assesses participants’ anxiety levels towards mathematics in different scenarios on a 5-point Likert scale, with higher scores indicating greater math anxiety.
- Academic Buoyancy: will be measured using the Academic Buoyancy Scale (Martin & Marsh, 2008). The 4-item questionnaire measures students’ ability to cope with everyday academic challenges using a 7-point Likert scale, with higher scores indicating greater academic buoyancy.
- Math avoidance: will be assessed using an adapted version of the Math Avoidance Scale (Conlon et al., 2022). The questionnaire measure participants’ avoidant thoughts or behaviours in 10 math-related situations on a 4-point Likert scale, with higher scores indicating greater math avoidance.
- Math Self-efficacy: will be assessed using an adapted version of Harter’s questionnaire (2012), measuring scholastic self-competence in the mathematical domain. The final 10-item version measure uses a 4-point Likert scale, with higher scores indicating greater math self-competence.

State-level math attitudes will be assessed using the following self-report questionnaires administered in close temporal proximity to task performance.

- State Worry: will be assessed using three adapted items from Mammarella et al. (2023). Items measure participants’ worry levels during the math task on a 4-point Likert scale, with higher scores indicating greater worry.

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- State Perceived Self-Competence: will be assessed using three adapted items from Mammarella et al. (2023). Items measure participants' perceived self-competence during the math task on a 4-point Likert scale, with higher scores indicating greater perceived self-competence.

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Indices

For screening measures, a total math score will be calculated based on accuracy in the math task at T0. Sum scores will be used for all trait and state questionnaires. Math behavioural choices will be coded at the item level as described in the "Measured variables" section.

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Analysis Plan

Statistical models

This section provides a brief overview of the data analysis steps and modelling approach. The full analysis pipeline, including data preparation, model specifications, and hypothesis tests, is implemented in the attached R script "Planned_analysis.R", which is applied to a simulated dataset ("DataSimulated.csv").

Data preparation

Data will be imported and pre-processed to include trials from T1, which constitute the outcome of interest. Measures of math performance and math attitudes assessed at T0, including both trait- and state-level variables, as well as additional trait-level measures collected at Session 2 (T1), will be merged with the behavioural data and included as predictors or covariates. For each item at T1, a single outcome variable (choice) will be derived, indicating the behavioural option selected by the student (skipping, help-seeking, self-reliance). All continuous predictors will be standardized (z-scores).

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Statistical modelling and hypothesis testing

To address RQ1, two competing multilevel models will be fitted using the brms package: 1) a cumulative logit mixed-effects model assuming an ordinal structure of behavioural choices (skipping < help-seeking < self-reliance); 2) a multinomial logit mixed-effects model treating behavioural choices as categorical. Models fit will be compared using leave-one-out cross-validation (LOO; Gelfand et al., 1992; Vehtari et al., 2015) and Watanabe-Akaike information criterion (WAIC; Watanabe, 2010) (see "Planned Analysis.R").

To address RQ2, directional hypothesis tests will be conducted within the selected multilevel model with all primary predictors (trait academic buoyancy, trait avoidance, state worry, state self-competence) estimated while controlling for the specified covariates (trait math anxiety, trait self-competence, prior math ability at T0, gender). Directional hypotheses will test whether posterior mean coefficients differ significantly from zero ($\alpha = 0.05$, one-tailed normal tests on posterior samples). For the ordinal model, single coefficients reflect monotonic effects; for the categorical model, planned contrasts compare choice probabilities (self vs. skip; help vs. skip; self vs. help).

Transformations

No data

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Inference criteria

To address RQ1, the model with higher ELPD under Pareto-smoothed LOO-CV (primary index) or lower WAIC (secondary index) will be selected as best-fitting (see "PlannedAnalysis.R").

If the two indices strongly disagree, model choice for subsequent analyses will be based on LOO, which is more robust than WAIC in finite samples with weak priors (Vehtari et al., 2017), while the answer to RQ1 regarding the ordinal vs. categorical nature of the dependent variable will be treated as inconclusive due to insufficient evidence.

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To address RQ2, effects will be deemed statistically significant if 95% highest density intervals (HDIs) exclude zero or if one-tailed p-values from normal tests on posterior means are < 0.05 (prior to multiple comparison correction). Directional expectations are specified a priori: positive coefficients expected for trait academic buoyancy and state self-competence (reflecting a higher probability of self-reliant behaviour relative to help-seeking and skipping), and negative for trait avoidance and state worry (reflecting a higher probability of skipping relative to help-seeking and self-reliant behaviour).

For the selected best-fitting model (ordinal or categorical) we will conduct four primary comparisons, corresponding to the effects of trait academic buoyancy, trait avoidance, state worry, and state self-competence. These tests will be evaluated separately and subsequently adjusted for multiple comparison correction. Since RQ2 tests all four predictors simultaneously, p-values (from normal tests on posterior means) will be corrected using the False Discovery Rate (FDR) procedure across the four comparisons.

Data exclusion

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Missing data

We expect a low rate of missing data for the questionnaire responses, as participants are forced to answer every item, with the task being administered on a computer. However, math trials in session 2 (32 items) may exhibit higher apparent missingness due to the explicit skip option alongside help-seeking and self-reliant choices. Skipped trials will not be treated as missing but will be retained as distinct behavioural category ("choice"=skip) in the outcome variable, with task accuracy coded as 0.

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Exploratory analysis

To inform future research, exploratory analyses may examine whether demographic characteristics are associated with the behavioural outcomes of interest or moderate the observed effects. These analyses are intended to be descriptive and hypothesis-generating and will be clearly distinguished from the preregistered confirmatory analyses.

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Other

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Other

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